

3-Day Data Collection Pipeline for a Journal-Level Survey

Object Counting Across Modalities: Google Scholar Seeds -> Claude Code/Codex Expansion -> 500+ Organized Papers

Purpose of this guide

This document is a practical operating plan for collecting more than 500 high-quality candidate papers for a journal survey on object counting across modalities. The goal is not to read 500 papers deeply in three days. The goal is to build a defensible corpus: a large candidate library, a clean literature matrix, a relevance-screened core set, and a transparent grouping structure for writing the survey.

Important rule

Use Google Scholar for manual seed discovery and verification. Do not build a scraper that repeatedly queries Google Scholar. Use Claude Code or Codex to automate organization and expansion through appropriate scholarly APIs such as Semantic Scholar, OpenAlex, Crossref, arXiv, PubMed when relevant, and exported BibTeX/CSV files.

1. Final Outputs After 3 Days

Output	What it should contain
master_papers_raw.csv / .xlsx	All raw papers collected from Scholar exports and API expansion, including duplicates. Target: 900-1,500 raw records.
literature_matrix_clean.xlsx	Deduplicated and screened working corpus. Target: 500-700 relevant candidate papers.
core_reading_set.xlsx	High-priority papers for deep reading. Target: 80-150 papers.
references.bib	BibTeX file for LaTeX writing. Include DOI/URL where available.
taxonomy_groups.md	Thematic grouping of papers by modality, method, prompt type, output, dataset, and application.
screening_log.xlsx	Transparent inclusion/exclusion decisions. Useful for journal reviewers.
survey_outline.md or .tex	A first survey outline generated from the organized literature matrix.

2. Three-Day Timeline Overview

Day	Main goal	Expected result
Day 1	Google Scholar seed collection	Build taxonomy, run controlled Scholar searches, manually collect 150-250 strong seed papers across modalities and datasets.
Day 2	Claude Code/Codex expansion	Use APIs to expand seeds through citations, references, related papers, venue searches, and keyword searches. Deduplicate to reach 500+ candidates.
Day 3	Screen, group, and prepare writing corpus	Score relevance, group into survey taxonomy, produce core set, BibTeX, PRISMA-style counts, and a writing outline.

3. Day 1 - Google Scholar Seed Collection

Day 1 target

Collect 150-250 seed papers manually from Google Scholar. These are not the final 500 papers. They are the high-quality starting points that Claude Code/Codex will expand using APIs on Day 2.

3.1 Create the seed spreadsheet before searching

Create a file named seed_papers_google_scholar.xlsx with these columns:

Column 1	Column 2	Column 3	Column 4
paper_id	title	authors	year

Column 1	Column 2	Column 3	Column 4
venue	modality	task_type	input_type
output_type	dataset	query_used	scholar_rank
bibtex_saved	pdf_saved	priority	notes

3.2 Search strategy: modality-first, not topic-only

Bucket	Google Scholar queries	Seed target
Survey / taxonomy	"object counting" "survey"; "visual counting" survey; "class-agnostic counting" survey; "few-shot object counting" survey	15-25
Image counting - general	"object counting" "computer vision"; "visual object counting"; "dense object counting"; "object counting and detection"	25-40
Few-shot / exemplar-based image counting	"few-shot object counting"; "exemplar-based object counting"; "class-agnostic object counting"; "FSC-147" "object counting"	40-60
Zero-shot / open-vocabulary image counting	"zero-shot object counting"; "open-vocabulary object counting"; "text-guided object counting"; "vision-language" "object counting"	35-50
Foundation-model counting	"object counting" "foundation models"; "object counting" "SAM"; "object counting" "CLIP"; "object counting" "vision language model"	20-35
Video counting	"video object counting"; "object counting in videos"; "open-world object counting in videos"; "unique instance counting" video; "CountVid"; "VideoCount"	35-50
Tracking-related counting	"video counting" tracking; "object counting" re-identification video; "tracking-based counting"; "TAO-Count"; "MOT20-Count"	20-35
Other modalities	"3D object counting"; "object counting" "point cloud"; "RGB-D object counting"; "event camera" object counting; "thermal" object counting	20-35
Domain modalities	"remote sensing" "object counting"; "aerial image" "object counting"; "cell counting" "computer vision"; "microscopy" "object counting"; "UAV" object counting	40-60
Datasets / benchmarks	"FSC-147"; "FSCD-147"; "CountBench" object counting; "OmniCount"; "CARPK" counting; "ShanghaiTech" counting; "TAO-Count"	30-50

How to search each query

For each query, inspect the top 20 results sorted by relevance and the top 10 results sorted by date. Save only papers that are directly relevant, highly cited, recent, dataset-defining, method-defining, or clearly useful for an application section. Use "Cited by" and "Related articles" only for the strongest papers in each bucket.

3.3 Scholar collection rules for quality

Decision	Rule
Include	Paper introduces a counting method, dataset, benchmark, evaluation protocol, survey taxonomy, or strong application case.
Include	Paper is about image, video, 3D, point cloud, RGB-D, aerial/remote sensing, microscopy, thermal, event-camera, or scientific visual counting.
Include	Paper is highly cited, very recent, published in a strong venue, or directly related to zero-shot, few-shot, open-vocabulary, class-agnostic, or video counting.
Exclude	Paper only mentions counting in passing but is mainly generic object detection, segmentation, or tracking.
Exclude	Paper is unrelated to visual objects, images, videos, or visual modalities.
Exclude	Paper has no accessible metadata and no clear relevance to the survey taxonomy.

3.4 Day 1 end-of-day checklist

- 150-250 seed papers saved in seed_papers_google_scholar.xlsx.

- At least 10-20 survey/taxonomy/dataset papers identified.
- Each paper has modality and task_type filled in.
- BibTeX copied for the strongest papers where available.
- Queries used are recorded, so the search process is reproducible.
- A folder exists for PDFs: papers_pdf/image, papers_pdf/video, papers_pdf/other_modalities, papers_pdf/datasets.

4. Day 2 - Claude Code/Codex Expansion to 500+ Papers

Day 2 target

Use Claude Code or Codex to turn the 150-250 Scholar seed papers into a larger corpus of 900-1,500 raw records, then deduplicate and filter into 500-700 relevant candidate papers.

4.1 Folder structure to create

```
object_counting_survey_collection/
  data/
    seed_papers_google_scholar.xlsx
    raw_api_results/
    master_papers_raw.csv
    literature_matrix_clean.xlsx
    screening_log.xlsx
  scripts/
    collect_openalex.py
    collect_semantic_scholar.py
    collect_crossref.py
    deduplicate_papers.py
    score_and_group_papers.py
    export_bibtex.py
  outputs/
    references.bib
    taxonomy_groups.md
    core_reading_set.xlsx
    survey_outline.md
  papers_pdf/
    image/
    video/
    other_modalities/
    datasets/
```

4.2 Expansion sources

Source	Use in the pipeline
Semantic Scholar	Paper metadata, citations, references, related works, abstracts, citation counts when available.
OpenAlex	Large scholarly metadata graph; useful for works, concepts, venues, authors, years, and related terms.
Crossref	DOI validation and publisher metadata.
arXiv	Recent preprints, especially for open-vocabulary, foundation-model, and video counting work.
PubMed / domain APIs	Optional for microscopy, cell counting, biomedical imaging, or healthcare applications.
Manual PDFs and BibTeX	Use exported Scholar/BibTeX records and downloaded PDFs only where allowed and accessible.

4.3 Prompt 1 - Ask Claude Code/Codex to build the pipeline

You are helping me collect literature for a journal-level survey paper titled:

"Object Counting Across Modalities: A Survey of Image, Video, Open-Vocabulary, and Emerging Visual Counting Methods."

I have a seed file at data/seed_papers_google_scholar.xlsx with columns:

title, authors, year, venue, modality, task_type, input_type, output_type, dataset, query_used, priority, notes.

Create a Python literature collection pipeline that:

1. Reads the seed spreadsheet.
2. Queries Semantic Scholar, OpenAlex, Crossref, and arXiv using paper titles, DOIs, and keywords.

Object Counting Survey Data Collection Pipeline

3. Expands each seed paper by collecting:
 - references
 - citing papers
 - related papers
 - same-venue papers with matching keywords
 - papers matching the modality-specific query bank
4. Saves all raw records to data/master_papers_raw.csv.
5. Deduplicates by DOI, arXiv ID, normalized title, and fuzzy title matching.
6. Produces data/literature_matrix_clean.xlsx.
7. Exports outputs/references.bib.
8. Produces outputs/taxonomy_groups.md grouping papers by modality, method family, prompt type, output type, dataset, and application area.
9. Logs every API query and screening decision for reproducibility.

Do not scrape Google Scholar. Use only the seed file exported from manual Scholar searches and legitimate APIs.

4.4 Prompt 2 - Query bank for API expansion

Create a query_bank.yaml file for API searches with these buckets:

```
survey_taxonomy:
  - object counting survey
  - visual counting survey
  - class agnostic counting survey
image_counting:
  - few-shot object counting
  - exemplar-based object counting
  - class-agnostic object counting
  - dense object counting
zero_shot_open_vocab:
  - zero-shot object counting
  - open-vocabulary object counting
  - text-guided object counting
  - vision-language model object counting
foundation_models:
  - object counting foundation models
  - SAM object counting
  - CLIP object counting
video_counting:
  - video object counting
  - open-world object counting in videos
  - unique instance counting video
  - tracking based object counting video
other_modalities:
  - 3D object counting
  - point cloud object counting
  - RGB-D object counting
  - event camera object counting
  - thermal object counting
applications:
  - remote sensing object counting
  - aerial image object counting
  - UAV object counting
  - microscopy cell counting
  - traffic object counting
  - wildlife object counting
benchmarks:
  - FSC-147 object counting
  - FSCD-147 object counting
  - CountBench object counting
  - OmniCount object counting
  - CARPK counting
  - ShanghaiTech counting
  - TA0-Count
  - VideoCount
```

For each query, retrieve up to 100 results when supported by the API, prioritize papers from 2015 onward, but keep older foundational papers if highly cited or dataset-defining.

4.5 Deduplication and minimum metadata requirements

Step	Rule
Dedup keys	DOI, arXiv ID, Semantic Scholar paper ID, OpenAlex work ID, normalized title, fuzzy title similarity.
Keep one best record	Prefer record with DOI, abstract, venue, year, citation count, PDF/OpenAccess URL, and BibTeX fields.
Minimum required fields	title, year, source, modality guess, relevance score, and reason_kept or reason_removed.
Flag incomplete records	Do not delete immediately. Mark as metadata_incomplete and review if title appears important.
Quality threshold	After deduplication, keep 500-700 papers with score $\geq$ threshold, plus all dataset/survey/foundational papers even if older.

4.6 Prompt 3 - Deduplicate and score the corpus

Read data/master_papers_raw.csv and create data/literature_matrix_clean.xlsx.

Deduplicate papers using:

- DOI exact match
- arXiv ID exact match
- Semantic Scholar/OpenAlex IDs
- normalized title exact match
- fuzzy title similarity above 92%

For each remaining paper, assign:

- modality: image, video, 3D, point_cloud, RGB-D, remote_sensing, microscopy, thermal, event_camera, multimodal, other
- method_family: density_map, detection_based, regression_based, exemplar_based, few_shot, zero_shot, open_vocabulary, vision_language, segmentation_based, tracking_based, re_identification, foundation_model, dataset_benchmark, survey
- input_type: image, video, text_prompt, exemplar_boxes, exemplar_patches, points, masks, depth, point_cloud, multimodal
- output_type: count_only, boxes_plus_count, masks_plus_count, tracks_plus_unique_count, density_map, benchmark
- application_area: general, agriculture, traffic, wildlife, healthcare, microscopy, remote_sensing, manufacturing, crowd_counting, other
- relevance_score from 0 to 5
- tier: A_core, B_important, C_background, Exclude
- reason_kept_or_removed

Scoring guide:

- 5 = central method/dataset/survey for object counting
- 4 = directly relevant to one modality or task family
- 3 = useful background or application paper
- 2 = weakly related; keep only if needed
- 1 = mostly unrelated
- 0 = remove

Save:

- data/literature_matrix_clean.xlsx
- outputs/core_reading_set.xlsx containing all A_core and high B_important papers
- outputs/screening_summary.md with counts by source, year, modality, method_family, and tier.

5. Day 3 - Screening, Grouping, and Journal-Ready Organization

Day 3 target
Turn the 500+ candidate corpus into a clean, reviewable literature base. The final survey should not simply list papers; it should show a taxonomy, trends, datasets, gaps, and why the field needs the survey.

5.1 Tier system for 500+ papers

Tier	Target size	How to use
A - Core reading set	80-150	Read deeply. These define the survey argument, taxonomy, method comparisons, and major tables.
B - Important supporting set	150-250	Read abstracts/method/results. Use for trend analysis, related work breadth, and citations.
C - Background/application set	200-350	Use selectively for applications, historical coverage, and modality breadth.

Tier	Target size	How to use
Exclude	Any number	Remove from final matrix or keep in a separate excluded sheet with reason.

5.2 Recommended survey taxonomy

Grouping axis	Categories
By modality	Image counting; video counting; 3D/point-cloud/RGB-D; remote sensing/aerial/UAV; microscopy/medical/scientific imaging; thermal/event-camera; multimodal settings.
By supervision/task setting	Fully supervised, density-map, detection-based, few-shot, exemplar-based, class-agnostic, zero-shot, open-vocabulary, text-guided, foundation-model-based.
By input prompt	No prompt, class label, text prompt, exemplar boxes, exemplar patches, points, masks, image prompt, multimodal prompt.
By output	Count only, density map, boxes plus count, masks plus count, tracks plus unique count, benchmark/evaluation protocol.
By temporal reasoning	Single-image count, frame-level video count, unique-instance video count, tracking-based count, re-identification-based count, occlusion/reappearance handling.
By dataset/benchmark	FSC-147/FSCD-147, CountBench, OmniCount, CARPK, ShanghaiTech, TAO-Count/VideoCount, domain-specific counting datasets.
By application	Agriculture, traffic analysis, wildlife monitoring, healthcare/cell counting, manufacturing/industrial inspection, crowd counting, remote sensing.

5.3 Prompt 4 - Generate grouped summaries and survey outline

Using data/literature_matrix_clean.xlsx, generate:

- outputs/taxonomy_groups.md
For each group, list:
 - representative papers
 - earliest/foundational papers
 - most recent papers
 - datasets used
 - common metrics
 - limitations and open gaps
- outputs/survey_outline.md
Suggested structure:
 - Introduction and motivation
 - Scope and inclusion criteria
 - Taxonomy of object counting across modalities
 - Image object counting methods
 - Few-shot, class-agnostic, zero-shot, and open-vocabulary counting
 - Video object counting and unique-instance counting
 - Emerging modalities: 3D, point cloud, RGB-D, remote sensing, microscopy, thermal, event-camera
 - Datasets, benchmarks, and evaluation metrics
 - Applications
 - Trends, gaps, and future directions
 - Conclusion
- outputs/tables_to_create.md
Propose the tables and figures needed for a journal survey, including:
 - taxonomy table
 - dataset table
 - method comparison table
 - modality comparison table
 - evaluation metrics table
 - timeline figure
 - future directions table
- outputs/core_reading_plan.md
Organize the A-tier papers into a reading order for writing the survey.

5.4 Prompt 5 - Produce a PRISMA-style screening log

Create a PRISMA-style screening summary from the files in data/.
Report:

- number of Google Scholar seed papers collected manually
- number of records retrieved from Semantic Scholar
- number of records retrieved from OpenAlex
- number of records retrieved from Crossref
- number of records retrieved from arXiv
- number of raw records before deduplication
- number removed as duplicates
- number screened by title/abstract
- number excluded and reasons
- number retained in final 500+ candidate corpus
- number in A_core, B_important, and C_background tiers

Save the result as outputs/screening_summary.md and data/screening_log.xlsx.

6. Quality Control Rules for Journal Acceptance

Quality rule	What to do
Reproducibility	Record every query, source, date searched, and inclusion/exclusion reason.
Coverage	Ensure the final corpus covers images, videos, and at least three emerging/specialized modalities.
Balance	Do not let crowd counting or generic detection dominate unless explicitly part of the survey scope.
Recency	Include strong recent papers from 2022-2026, especially foundation-model, zero-shot, open-vocabulary, and video counting papers.
Foundations	Keep older foundational papers if they introduced a dataset, metric, or method family.
Taxonomy	Every included paper should map to at least one modality and one method family.
Evidence	Use tables and counts to demonstrate that the survey corpus is systematic, not random.
Manual review	Do not trust automatic labels blindly. Manually review all A-tier papers and questionable B-tier papers.

7. Suggested 3-Day Work Schedule

Time	Task
Day 1 morning	Create spreadsheet, folders, and taxonomy. Run survey/dataset Scholar searches.
Day 1 afternoon	Run image, zero-shot, open-vocabulary, and foundation-model Scholar searches.
Day 1 evening	Run video, other-modality, and application searches. Clean the seed spreadsheet.
Day 2 morning	Run Claude Code/Codex Prompt 1 and Prompt 2. Generate API collection scripts.
Day 2 afternoon	Collect raw API results. Expand through references/citations/related works.
Day 2 evening	Deduplicate and score. Aim for 500-700 relevant candidate papers.
Day 3 morning	Manually inspect A-tier and uncertain B-tier papers. Fix group labels.
Day 3 afternoon	Generate taxonomy_groups.md, core_reading_set.xlsx, references.bib, and screening summary.
Day 3 evening	Generate survey outline, tables-to-create list, and journal-ready documentation of the search process.

8. Final Checklist Before Writing the Survey

- At least 500 deduplicated candidate papers are retained in literature_matrix_clean.xlsx.
- At least 80-150 A-tier papers are identified for deep reading.
- Every paper has modality, method_family, input_type, output_type, dataset, and relevance_score fields.
- references.bib compiles without obvious duplicate keys.
- screening_log.xlsx explains why papers were kept or excluded.
- taxonomy_groups.md shows clear sections for images, videos, and other modalities.
- survey_outline.md is aligned with the taxonomy and not just chronological listing.

- A table/figure plan exists for the journal manuscript.
- The process avoids automated scraping of Google Scholar and uses APIs for scalable expansion.